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From Greedy to Genius: Understanding Decoding Strategies in Large Language Models

How Large Language Models decide what to say next and how temperature, top-p, and other decoding parameters let you tune their creativity and control.

5 min read · Oct 21, 2025



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Every time an LLM generates text, it doesn't write sentences; it predicts one token at a time. The magic lies in how it chooses that next token.

This piece explains the decoding strategies behind text generation, from greedy decoding to sampling, and how tweaking parameters such as temperature, top-k and top-p can transform your model's personality.

If you have ever tried generating text with a Large Language Model (LLM), whether through an API, a chatbot, or an in-house deployment, you have probably noticed something curious.

Sometimes the model sounds brilliant and almost human. At other times, it rambles, repeats itself, or becomes strangely mechanical.

You might adjust parameters such as temperature or top-p, hoping to make it more creative, but what is actually happening beneath the surface when you do that?

This article explores that question. It explains how a Large Language Model decides what to say next, what decoding strategies really mean, and how you can fine-tune these parameters to balance creativity and control, transforming raw predictions into purposeful, human-like writing.

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How an LLM Thinks (One Token at a Time)

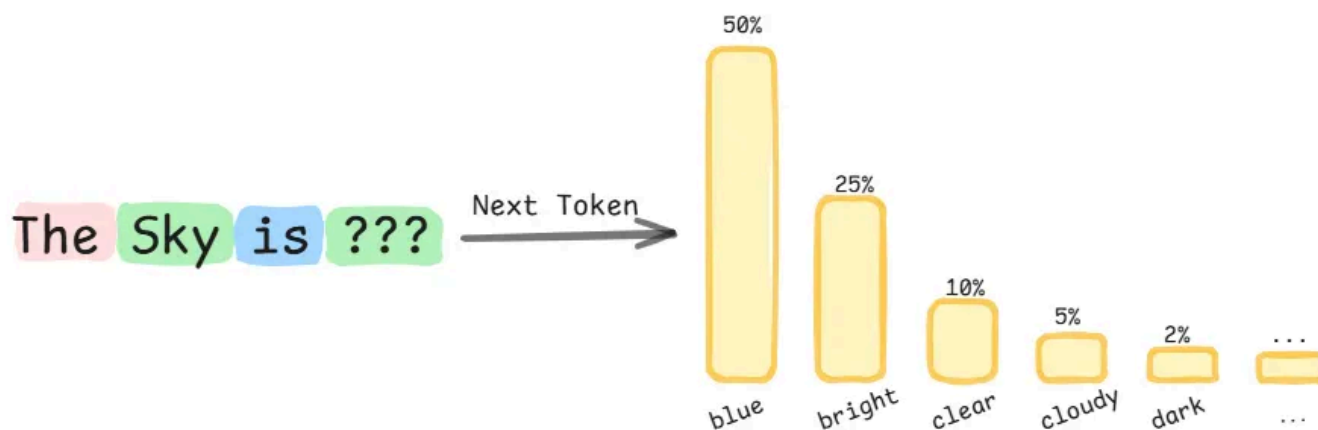
Before we explore the knobs and levers, let us look more closely at the generation process itself.

A large language model does not think in full sentences; it generates text one token at a time.

At each step, it:

- Reads your input and its own previous outputs (the context).
- Predicts which token might come next, based on probabilities.
- Assigns a score to every possible token in its vocabulary.
- Selects one token to output.

The final step, deciding which token to choose, is what decoding strategies control. Think of it as the model's decision-making layer, where probabilities become actual words.



Decoding Strategies: Predictable vs. Creative

There are two main philosophies behind decoding; one values **certainty**, the other **variety**.

1. Greedy Decoding — The Predictable Path

Greedy decoding is straightforward: at each step, the model chooses the token with the **highest probability**.

It's simple, deterministic, and fast. Run it twice, and you'll always get the same result.

The trade-off? It's often dull and repetitive.

Greedy decoding is so focused on the “most likely” next word that it forgets the broader picture — the flow and creativity of natural language.

Example: “The sky blue.”

What you probably wanted: “The sky is cloudy”

2. Sampling — The Creative Route

Sampling gives the model some freedom. Instead of always taking the top token, it **samples** from the entire probability distribution, meaning less likely words sometimes get chosen.

This adds **randomness** and **diversity**, producing more engaging, natural text. But it also means results are *non-deterministic*: the same input can yield slightly different outputs.

In most frameworks, this is enabled with `do_sample=True`.

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Sampling by itself is only the start. To control it effectively, you use several parameters that shape how much freedom or discipline the model has.

The Four Knobs of Text Generation

Here's where the engineering craft comes in.

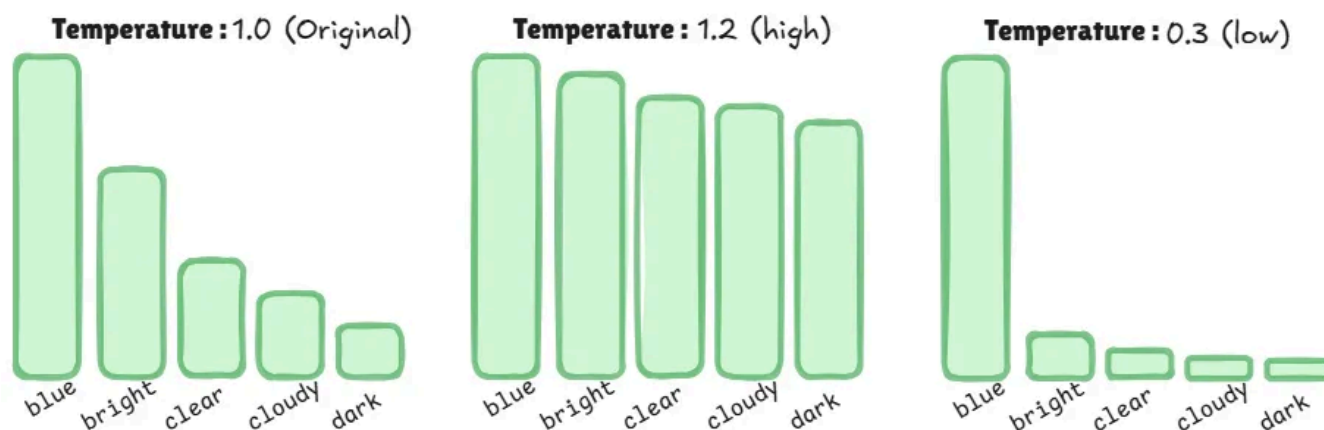
Each of the following parameters affects how the model “thinks” when sampling its next token.

1. Temperature: Controlling the Chaos

Temperature adjusts how random or focused the output is.

- **Low temperature (≈ 0.1 – 0.5):** Conservative, deterministic text. Ideal for structured or factual responses.
- **High temperature (≈ 0.8 – 1.2):** Adds flair and unpredictability — great for brainstorming or creative writing.
- **Temperature = 0:** Equivalent to greedy decoding.

Think of temperature as like a dial that changes the shape of probability distribution by rescaling these probabilities and in this way influencing which token is chosen.

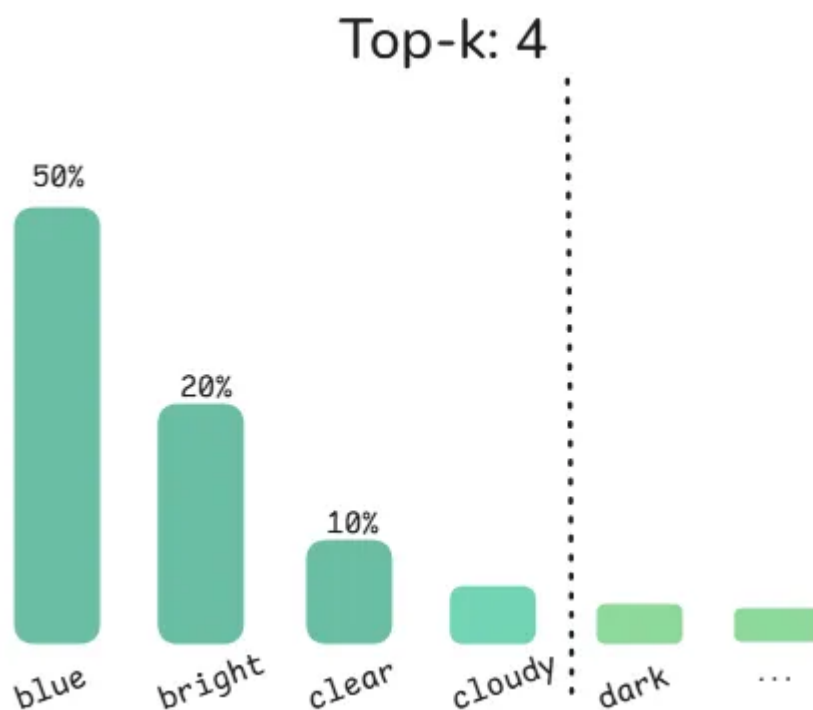


2. Top-K Sampling: Limiting the Options

Here, the model considers only the **K most likely tokens** before picking one at random.

- **Small K (e.g., 10):** Predictable, focused outputs.
- **Large K (e.g., 100+):** Broader vocabulary, more creativity.

Top-K acts like a shortlist — it doesn't change the ranking of tokens, just the number of options on the table.



3. Top-P (Nucleus) Sampling: Context-Sensitive Filtering

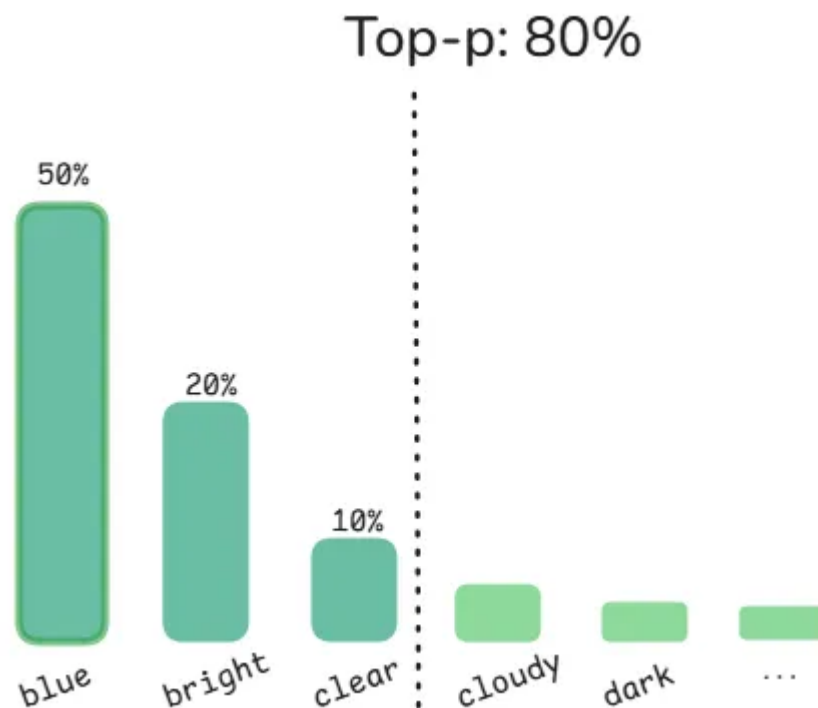
Top-P, or *nucleus sampling*, takes a more adaptive approach.

Instead of using a fixed number of tokens, it includes as many as needed to reach a **cumulative probability threshold (p)**.

If $p = 0.8$, the model considers just enough tokens whose combined probability sums to 80%.

When the model is confident, that might be a handful of tokens; when it's uncertain, the pool expands.

Top-P adapts to context, producing text that's often smoother and more contextually aware than fixed Top-K.



4. Repetition Penalty: Keeping Things Fresh

LLMs can sometimes fall into loops — “The sky is blue. The sky is blue. The sky is blue.” To prevent that, we apply a **repetition penalty**, which discourages the reuse of previously generated tokens.

- **Typical value:** 1.2
- **Higher values (1.5–2.0):** Reduce repetition but can make phrasing less natural.

It's a small tweak that dramatically improves flow and readability.

Top-K vs. Top-P: What's the Real Difference?

Strategy	Based on	Description	Behaviour
Top-K	Token rank	Chooses the K most likely tokens	Fixed shortlist
Top-P	Probability mass	Includes tokens until the cumulative probability equals p	Adaptive shortlist

Top-K sets a fixed cut-off regardless of context, while Top-P adjusts dynamically based on confidence, making it more nuanced and context aware.

How to Tune These Knobs

Goal	Parameter	Direction
More Creative, Diverse Output	Temperature ↑	Encourages less probable words
	Top-P ↑ (0.9–0.95)	Broadens the sampling pool
	Top-K ↑ (100+)	Increases variety
More Consistent, Deterministic Output	Temperature ↓	Makes predictions stable
	Top-P ↓ (0.3–0.5)	Narrows the candidate pool
	Top-K ↓ (10–20)	Restricts to highly probable tokens
Reduce Repetition	Repetition Penalty ↑ (1.2–2.0)	Suppresses repeated tokens

*There's no universal setting. The best models are tuned by experimentation, testing what "feels right" for your task. These parameters are what shape the **model's personality**.*

- Turn them down and you get a careful, precise assistant.

- Turn them up, and you get something more imaginative and expressive.

When we talk about Large Language Models being creative or predictable, we are really talking about how they decode and how they choose each word. Once you

understand these **decoding strategies**, you move from using a model to shaping it. You begin to see the output not as random chance but as something you can design, adjust, and guide. Large Language Models are not simply text predictors; *they are probabilistic storytellers*. The art of working with them lies in **tuning the balance** between *predictability* and *creativity*, depending on what you are building.

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Muhammad Azam Mehr Ghulam is a Java Technical Lead here at [Version 1](#).

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